

Runs I & II @ 3X to 1.5X FeSO₄ vs. usual FeSO₄

Run #1 @ 1% NVR	Run #2 @ 1% NVR	Run #3 @ 1% NVR	Run #4 @ 1% NVR	Run #5 @ 1% NVR	Run #6 @ 1% NVR
G 2.186	G 2.180	G 2.147	G 1.731	G 1.998	G 1.662
T 1.327	T 1.393	T 1.367	T 1.367	T 1.369	T 1.369
N 1.323	N 0.619	N 0.780	N 0.364	N 0.629	N 0.293
46.8%					
Add 110 gms water to 149 gms product					
46.6%					

Run #1 @ 1% NVR

Run #1 @ 1% NVR	Run #2 @ 1% NVR	Run #3 @ 1% NVR	Run #4 @ 1% NVR	Run #5 @ 1% NVR	Run #6 @ 1% NVR
G 2.007	G 1.667	G 1.919	G 1.618	G 1.995	G 1.655
T 1.371	T 1.371	T 1.357	T 1.357	T 1.361	T 1.361
N 0.636	N 0.296	N 0.562	N 0.261	N 0.624	N 0.294
46.5%					
Add 75 gms water to 1055 gms product					
47.1%					

Water #1

G 241.3
T 131.3
N 110.0 g

Water #2

G 206.8
T 131.8
N 75.0 g

I. 67	Run #1 @ 1% NVR	Run #2 @ 1% NVR	Run #3 @ 1% NVR	Run #4 @ 1% NVR	Run #5 @ 1% NVR
G 2.133	G 1.694	G 2.130	G 1.692	G 2.245	G 1.750
T 1.351	T 1.351	T 1.348	T 1.348	T 1.350	T 1.350
N 0.782	N 0.343	N 0.782	N 0.344	N 0.895	N 0.400
43.9%					
44.0%					

II. 47	Run #1 @ 1% NVR	Run #2 @ 1% NVR	Run #3 @ 1% NVR	Run #4 @ 1% NVR	Run #5 @ 1% NVR
G 2.129	G 1.701	G 2.264	G 1.765	G 2.183	G 1.739
T 1.365	T 1.365	T 1.371	T 1.371	T 1.368	T 1.368
N 0.764	N 0.336	N 0.893	N 0.394	N 0.815	N 0.360
44.0%					
44.1%					
44.2%					

Continued to page 4

A. L. Craig

1/31/85

Read and Understood By

Continued on Page

Signed

Date

Signed

Date

Run #1 @ 2X Fe SO₄ : Added 35 gms more water

	(17)	@ 30 min	(17)	@	(17)	@
G	2.336	1.794	2.040	1.665	2.119	1.783
T	1.389	1.385	1.383	1.383	1.389	1.389
N	0.951	0.409	0.658	0.283	0.729	0.314 g
		43.0%		43.0%		43.1%

Run #16 @ 1.5X Fe SO₄ : Added 26 gms more water

	(17)	@ 30 min	(17)	@ 30 min	(17)	@ 30 min
G	2.113	1.693	2.139	1.696	2.093	1.688
T	1.391	1.391	1.373	1.373	1.370	1.370
N	0.732	0.312	0.756	0.328	0.723	0.310 g
		42.6%		42.7%		42.9%

Continued on Page

R L Craig

Signed

2/1/85

Date

Read and Understood By

Signed

Date

Adjustment of % NVR: Run No. III

	(1) @ 30 min	(2) @ 30 min	(3) @ 30 min	(4) @ 30 min	Residual
G	2.055	1.702	2.121	1.732	1.966
T	1.382	1.382	1.377	1.377	1.381
N	0.673	0.320	0.744	0.355	0.585
	47.5%		47.4%		47.0%

Avg: 47.4

$$1216 \text{ (0.474)} / 0.431 = 1337.3 ; 1337 - 1216 = 121 \text{ g water}$$

	(4) @ 30 min	(5) @ 30 min	(6) @ 30 min
G	2.125	2.228	2.246
T	1.366	1.369	1.369
N	0.759	0.859	0.882
	43.7%	43.9%	43.9%

H₂-Heat (15 min)

	@ 45 min
G	1.694
T	1.366
N	0.328
	43.2%

	@ 45 min
G	1.741
T	1.369
N	0.372
	43.3%

	@ 45 min
G	1.745
T	1.364
N	0.331
	43.2%

Avg: 43.2%

Continued on Page

R L Craig

Signed

2/1/85

Date

Read and Understood By

Signed

Date

Lab Run No. 12; % Solids adjustment

(1)	@ 30 min	(3)	@ 30 min	(5)	@ 30 min
G 1.988	1.663	2.001	1.670	1.963	1.653
T 1.372	1.372	1.373	1.373	1.313	1.373
N 0.616	0.391	0.638	0.391	0.590	0.379 gms
	47.2%		47.3%		47.3%

$1025(0.443)/0.442 = 1122 \text{ gms}$; $1122 - 1025 = 97 \text{ gms}$
 Added 97 gms water to 1025 gms ~~water~~ Neut. Polymer

(1)	@ 30 min	(3)	@ 30 min	(5)	@ 30 min
G 2.004	1.656	2.110	1.702	2.853	1.677
T 1.388	1.388	1.384	1.384	1.336	1.386
N 0.616	0.368	0.726	0.318	0.667	0.391 gms
	43.5%		43.8%		43.6%

$1120(0.436)/0.431 = 1133$; add 13 gms more water

(1)	@ 30 min	(3)	@ 30 min	(5)	@ 30 min
G 2.042	1.670	2.092	1.673	2.003	1.658
T 1.388	1.388	1.391	1.391	1.396	1.396
N 0.654	0.292	0.701	0.302	0.606	0.262 gms
	43.1%		43.1%		43.2%

Added 6 drops of caustic to bring pH 8.0 back.

Continued on Page

Read and Understood By

B L Craig 2/4/85

Signed

Date

Signed

Date